在定量社会科学研究中负责任地使用人工智能

社会研究方法课程单元

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研究方向：运用因果推断与计算方法评估数字信息技术的社会与政治影响

2026 年 11 月 5 日

本次课的目标

（……（计算技能）门槛降低……专业知识的重新分配……）

“对学生而言，问题不再是他们能否编程，而是他们能否审视、批判、调整并负责任地部署 AI 辅助方法，而不把便利误当作……”(Zhou et al., 2026)

the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

for competence.

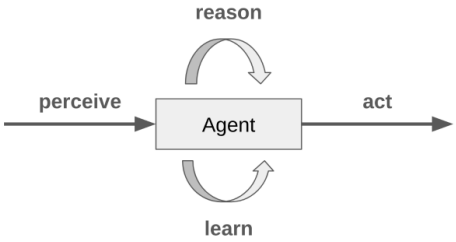
At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

来源：Zhou et al. (2026)

引言

1.1 什么是人工智能?

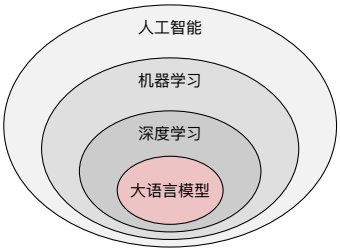
- 能够智能地行动的计算智能体 (Poole & Mackworth, 2023)
- 但什么是智能? ——见右图



来源: Stanford CS221, Percy Liang

1.1 什么是人工智能?

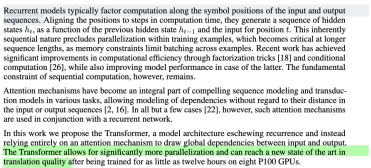
- 能够智能地行动的计算智能体 (Poole & Mackworth, 2023)
- 但什么是智能? ——见右图
- 更重要的是，请注意以下概念之间的区别 ……



1.2 大语言模型简史

- * 人工智能是一个已发展数十年的古老领域 (参见 Poole & Mackworth, 2023)
- * 前大模型时代，早期工具主要基于人类产生的痕迹数据进行分类、聚类或预测 (Zhou et al., 2026)。[a]
- * 大语言模型的诞生：Transformer (Vaswani et al., 2017)，一种可并行训练的模型结构，使大规模数据与大模型成为可能
- 后大模型时代：本课聚焦 2020 年后的大语言模型时代 (以 GPT-3 为标志； Xu et al., 2026) 及其对计算社会科学研究的意义；并简要演示基于 BERT 的文本分类

[a] 类似地，Percy Liang 以 2016 年、2018 年和当下划分三个时期，见 https://cs336.stanford.edu/lectures/?trace=lecture_01



Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

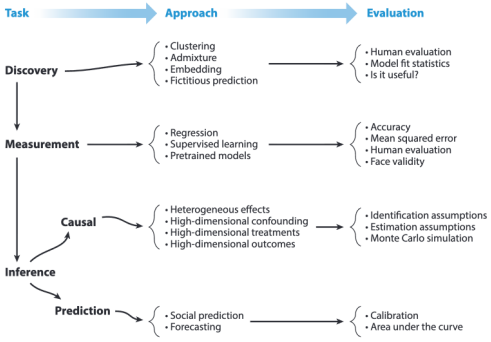
End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and

摘自 Vaswani et al. (2017)

前大模型时代的 AI

2.1 前大模型时代 AI 在社会科学中的应用

- 按照 Grimmer et al. (2021) 的分类，社会科学研究常借助机器学习完成多种任务：
  - 发现；例如文本分类 （教程一）
  - 测量；例如
  - 因果推断；例如
  - 预测；例如
- 如有兴趣，可选修汇丰商学院 FIN528 金融机器学习 / 高级机器学习



来源：Grimmer et al. (2021)（第 23 页）

TUT1

使用 RoBERTa 进行情感分类

大语言模型

3.1 大语言模型在计算社会科学中的角色

- 大语言模型（LLM）大致指参数量不少于 10 亿的语言模型（Percy Liang；如 GPT、Claude）[1]
- 一篇近期综述提出大语言模型在计算社会科学中的四种角色：研究主体、助手、合作者、评估者

Table 1 | Representative recent work leveraging LLMs in social scientific study, organized by the research roles they have

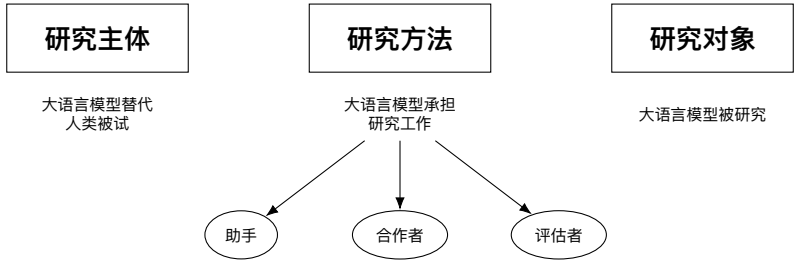
Research role	Description	Example works
Research subjects	Generating survey responses	Refs. 32,43,48
	Simulating behaviour sequences	Refs. 52,53,55
	Simulating social interaction	Ref. 61,63,65,67
Research assistants	Interviewing human participants	Refs. 44,73,74,76
	Supporting experiments	Refs. 81–83
	Analysing textual content	Refs. 85,93,99,102
	Mining social networks	Refs. 107–109
Research collaborators	Reviewing literature	Refs. 117,120,122
	Proposing hypotheses	Refs. 126–128
	Designing research protocols	Refs. 129,130
Research evaluators	Evaluating research protocols	Refs. 132,133,137,140
	Critiquing research outcomes	Refs. 141,142,144

来源：Xu et al. (2026)

[1]本课并非计算机课程，采用这一定义应无大碍。

3.1 大语言模型在计算社会科学中的角色

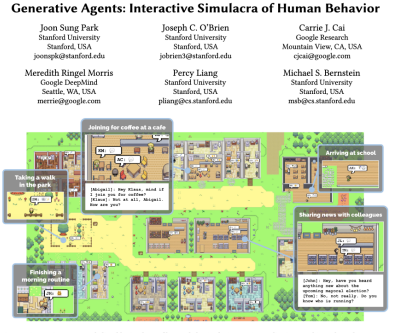
- 从更多视角出发，我们可以对大语言模型在计算社会科学中的角色提出更全面的分类；后续幻灯片将逐一展开



来源：四种角色（研究主体、助手、合作者、评估者）的分类见 Xu et al. (2026)；四个类别见 Hai Liang (2026)；两种角色（研究工具、研究对象）改编自 Zhou et al. (2026)

3.2 大语言模型作为研究主体：类别

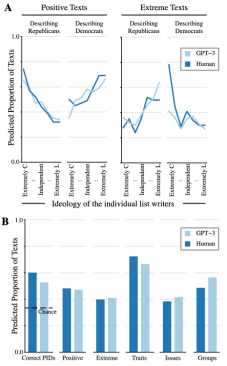
- 根据 Xu et al. (2026)，大语言模型可充当研究主体：
  - 生成问卷回答 (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hämäläinen et al., 2023; Y. Xie & Xie, 2025)
  - 模拟行为序列 (Park, O'Brien, et al., 2023; L. Wang et al., 2025; Yan et al., 2024)
  - 模拟社会互动 (Ashery et al., 2025; Gao et al., 2025; C. Xie et al., 2024; Zhang et al., 2024)



模拟小镇中的生成式智能体；Park, O'Brien, et al. (2023)

3.3 大语言模型作为研究主体：优势与不足

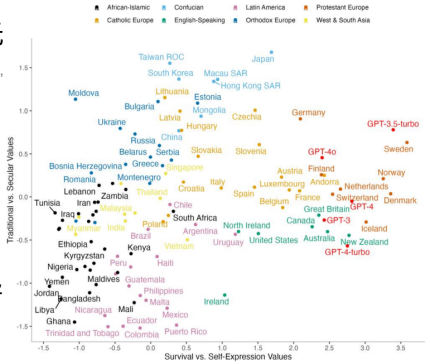
- **优势：**
  - 在问卷调查中成功复现人类回答 (Argyle, Busby, et al., 2023)
  - 产生可信的个体行为与涌现的社会行为 (Park, O'Brien, et al., 2023)
  - 复现社会规范的涌现 (Ashery et al., 2025)
  - 等等



GPT-3 与 Lucid 调查中人类回答的对比分析；Argyle, Busby, et al. (2023) 图 3

3.3 大语言模型作为研究主体：优势与不足

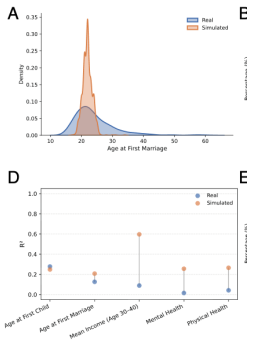
- **不足：**
  - 难以反映“难以触达的人群”(Bollen et al., 2026)；无法捕捉经验与文化层面的深度 (Pataranutaporn et al., 2025)
  - 体现特定的党派 (Santurkar et al., 2023) 或文化偏差 (Tao et al., 2024)
  - 接近均值但变异不足 (Bisbee et al., 2024)；在意见或态度研究中存在强烈且多变的偏差、方差偏低 (Boelaert et al., 2025)；容易同时误描并扁平化人口群体特征 (A. Wang et al., 2025)；并倾向于将异质性压缩为简化的类型结构 (Y. Xie & Xie, 2025)



大语言模型生成的价值观回答与不同国家相比存在偏离；
Tao et al. (2024) 图 1

3.3 大语言模型作为研究主体：优势与不足

- **不足：**
  - 难以反映“难以触达的人群”(Bollen et al., 2026)；无法捕捉经验与文化层面的深度 (Pataranutaporn et al., 2025)
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未能捕捉真实世界分布的离散程度；Y. Xie & Xie (2025) 图 3

3.4 大语言模型作为研究主体：评估

- 效度与信度仍是关注焦点 (Zhou et al., 2026)；请参阅叶老师此前的课程



FIGURE 5-2

An Analogy to Validity and Reliability. A good measurement technique should be both valid (measuring what it is intended to measure) and reliable (yielding a given measurement dependably).

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they are somewhat superior to those they study. For example, researchers often seek to uncover motivations that the social actors themselves are unaware of. You think you bought that new Turbo Tiger because of its high performance and good looks, but we know you're really trying to achieve a higher social status.

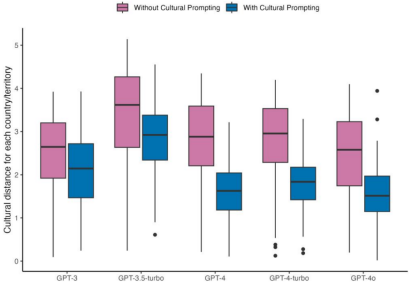
This implicit sense of superiority would fit comfortably with a totally positivistic approach (the biologist feels superior to the frog on the lab table), but it clashes with the more humanistic and typically qualitative approach taken by many social scientists. We'll explore this issue more deeply in Chapter 10. In seeking to understand they are escapes. The fact is that to commit oneself to a semiotic concept of culture and an interpretive approach to the study of it is to commit oneself to a view of ethnographic assertion as, to borrow W. B. Gallie's by now famous phrase, "essentially contestable." Anthropology, or at least interpretive anthropology, is a science whose progress is marked less by a perfection of consensus than by a refinement of debate. What gets better is the precision with which we vex each other. *(1973: 29)*

Ultimately, social researchers should look both to

效度与信度；见 Babbie (2021)

3.5 大语言模型作为研究主体：改进

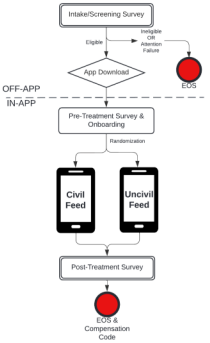
- 效度与信度仍是关注焦点 (Zhou et al., 2026)
- 三种改进策略
  - 对齐（使用前）：调整模型输出；主要由企业和计算机科学研究完成
  - 优化（使用中）：通过**微调或提示**策略提升特定任务的表现 (如 Kolluri et al., 2025; Tao et al., 2024; Y. Xie & Xie, 2025)
  - 校正（使用后）：事后修正；计量经济学与机器学习文献中的新兴方向 (如 Chernozhukov et al., 2018; Egami et al., 2024)



提示可缓解文化偏差；来源：Tao et al. (2024) 图 2

3.6 大语言模型作为研究方法：助手

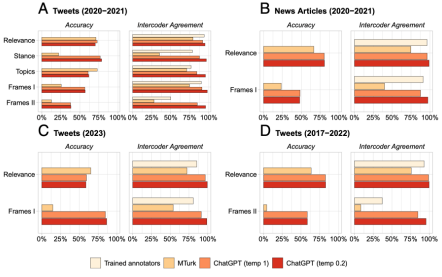
- 访谈人类参与者（质性研究）(Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- **支持实验研究** (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- 分析文本内容 (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- 挖掘社交网络 (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



随机实验中的合成传播环境；Allamong et al. (2026)

3.6 大语言模型作为研究方法：助手

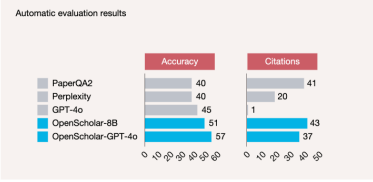
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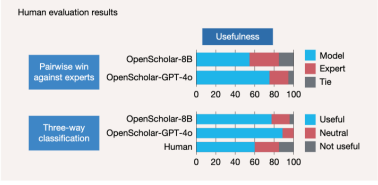


标注质量优于众包工作者；Gilardi et al. (2023)

3.7 大语言模型作为研究方法：合作者

- **文献综述** (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- 提出假设 (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- 设计研究方案 (Baek et al., 2025; Manning et al., 2024)

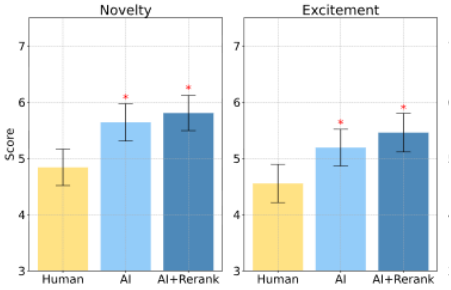




由大语言模型完成的博士水平文献综述系统；见 Asai et al. (2026)

3.7 大语言模型作为研究方法：合作者

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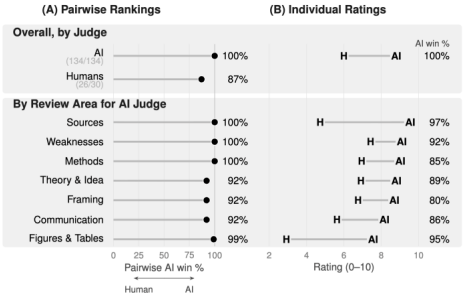
大语言模型生成的假设在新颖性与吸引力上更优；见 Si et al. (2024)

TUT2

使用大语言模型进行主题分类
别忘了效度检验！

3.8 大语言模型作为研究方法：评估者

- 评估研究方案 (Lin & Chen, 2023; Shankar et al., 2024; X. Wang et al., 2024; Zheng et al., 2023)
- **评议研究成果** (Ke et al., 2024; King & Worthington, 2026; M. Li et al., 2024; Liang et al., 2024)



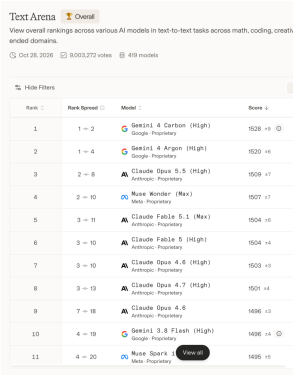
AI 评审全面优于人类评审；来源：(King & Worthington, 2026)；未经同行评议，但设计严谨

3.9 负责任地使用大语言模型：模型选择

- 将模型视为一个函数 $f(data, algo, object)$，各部分的差异会显著塑造其领域内能力（如更擅长编程或写作）
- 因此，我们可能需要根据不同模型的优势与局限，为不同任务选择不同模型

3.9 负责任地使用大语言模型：模型选择

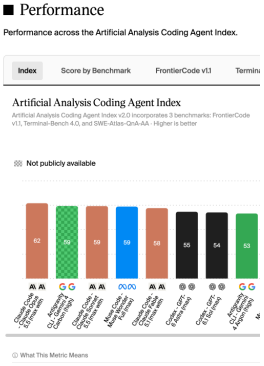
- 例如，Gemini Carbon 可能更擅长写作任务，而 Claude Opus 5.5 在工程任务上表现更好



来源：https://arena.ai/leaderboard/chat/text（2026 年 11 月 3 日）

3.9 负责任地使用大语言模型：模型选择

- 例如，Gemini Carbon 可能更擅长写作任务，而 Claude Opus 5.5 在工程任务上表现更好
- 总体而言，各模型厂商的商业化路径不同，面向工程任务优化商业模式的厂商可能优先发展编程能力（如 Anthropic）；使用某一模型前，请查看基准测试、回顾文献或进行预测试



来源：
https://artificialanalysis.ai/agents/coding-agents#coding-agents-index
(2026 年 11 月 3 日)

3.10 大语言模型作为研究对象

- “一种重塑媒介系统中生产、流通、阐释与权力的传播技术，引发了关于真实性、可及性以及社会不平等再生产的关键问题。”(Zhou et al., 2026)
- 例如，Liu and Zhang (2026) 评估了大语言模型生成的摘要如何塑造公众情绪，其处理变量即为模型本身。

RQ2. Whether and how do the emotions of AIGS influence sentiment in subsequent posts associated with the hashtag

To explore the effect of emotions in AIGS on the subsequent public sentiment, we used a Difference-in-Difference (DiD) design in natural experiments on Weibo. Before the DiD analysis, we conducted an event study plot on sentiment scores to examine the parallel assumption. As Figure 3 shows[2], the parallel trends assumption was met, indicating that there is no evidence of systematic platform-level selection bias between hashtags with and without the AIGS feature.

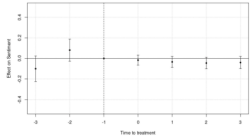


Figure 3: Event Study Plot on Sentiment Scores (Treatment versus Control)

Liu & Zhang (2026) 中作为研究对象的 AI 生成摘要（AIGS）

小测验

随堂问题：在线上小组中提出一个以大语言模型为研究对象的研究问题

局限性

4.1 局限性

1. **效度。**输出未必测量了我们想测量的内容
2. **可复现性。**闭源模型、静默更新与随机输出
3. **偏差。**训练数据对部分群体和语言代表性过高
4. **幻觉。**捏造事实与文献
5. **伦理。**隐私、知情同意以及向第三方传输数据
6. **过度依赖。**技能退化与未经审视的默认设置

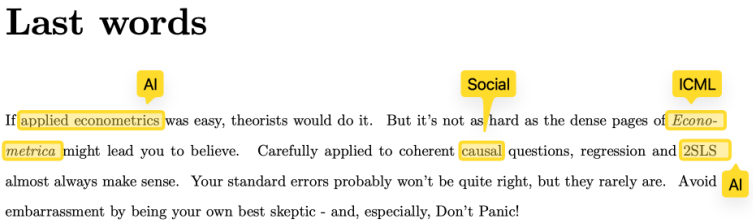
要点总结

5.1 要点总结

- AI 模型（包括大语言模型）在定量社会研究中可以充当研究主体、研究对象和研究方法
- 作为研究方法，应以负责任的方式使用，审慎考虑效度、可复现性、偏差、幻觉、伦理与过度依赖等问题
- 与其他方法一样，它们发展迅速（且可能越来越快），但对好问题的品味始终是研究（以及业界工作；相信我）的核心

5.1 要点总结

- 最后一句话：“……保持怀疑……但别慌！”(Angrist & Pischke, 2009)



Last words

AI Social ICML

If applied econometrics was easy, theorists would do it. But it's not as hard as the dense pages of *Econometrica* might lead you to believe. Carefully applied to coherent causal questions, regression and 2SLS almost always make sense. Your standard errors probably won't be quite right, but they rarely are. Avoid embarrassment by being your own best skeptic - and, especially, Don't Panic! AI

来源：Angrist & Pischke (2009)

资源

6.1 资源

- **为任务找到最合适的 AI 模型**
  - Artificial Analysis https://artificialanalysis.ai/
  - Arena.ai，AI 模型盲测排行榜 https://arena.ai/leaderboard/
- **发现稿件中的问题**
  - Refine，发表级技术审阅；已被美国经济学会（AEA）采用 https://www.refine.ink/
  - OpenAIReview，芝加哥大学 HAI 开源的多智能体稿件审阅方案 https://github.com/ChicagoHAI/OpenAIReview
- **复核参考文献与引用**
  - CiteVerify，“一款用于核验论文参考文献的本地工具”，https://github.com/cuihuash/CiteVerify
  - 与前沿模型对话也会有所帮助

6.1 资源

- **统一管理多个 AI 模型与服务**
  - macOS：magpie，Mac 本地 AI 路由管理工具 https://github.com/yetone/magpie
  - 其他平台：cc-switch，https://github.com/farion1231/cc-switch
- **用母语阅读科学文献**
  - PDFMathTranslate，以你的语言阅读科学 PDF
    https://github.com/PDFMathTranslate/PDFMathReader (Ouyang et al., 2025)

参考文献

7.1 参考文献 I

Allamong, M. B., Bail, C., Hillygus, D. S., & Volfovsky, A. (2026). The Causal Effects of Political Incivility in Social Media Discussions. Retrieved March 17, 2026, from https://osf.io/download/yaq29

Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton university press.

Argyle, L. P., Bail, C. A., Busby, E. C., Gubler, J. R., Howe, T., Rytting, C., Sorensen, T., & Wingate, D. (2023). Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences of the United States of America*, *120*(41), e2311627120. https://doi.org/10.1073/pnas.2311627120

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis*, *31*(3), 337–351. https://doi.org/10.1017/pan.2023.2

Asai, A., He, J., Shao, R., Shi, W., Singh, A., Chang, J. C., Lo, K., Soldaini, L., Feldman, S., D'Arcy, M., Wadden, D., Latzke, M., Sparks, J., Hwang, J. D., Kishore, V., Tian, M., Ji, P., Liu, S., Tong, H., … Hajishirzi, H. (2026). Synthesizing scientific literature with retrieval-augmented language models. *Nature*, *650*(8103), 857–863. https://doi.org/10.1038/s41586-025-10072-4

Ashery, A. F., Aiello, L. M., & Baronchelli, A. (2025). Emergent social conventions and collective bias in LLM populations. *Science Advances*, *11*(20), eadu9368. https://doi.org/10.1126/sciadv.adu9368

Babbie, E. R. (2021). *The practice of social research* (15th ed.). Cengage.

Baek, J., Jauhar, S. K., Cucerzan, S., & Hwang, S. J. (2025). ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models. In L. Chiruzzo, A. Ritter, & L. Wang (Eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 6709–6738). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.naacl-long.342

Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., & Larson, J. M. (2024). Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis*, *32*(4), 401–416. https://doi.org/10.1017/pan.2024.5

Boelaert, J., Coavoux, S., Ollion, É., Petev, I., & Präg, P. (2025). Machine Bias. How Do Generative Language Models Answer Opinion Polls?1. *Sociological Methods & Research*, *54*(3), 1156–1196. https://doi.org/10.1177/00491241251330582

Bollen, P., Higton, J., & Sands, M. (2026). Nationally Representative, Locally Misaligned: The Biases of Generative Artificial Intelligence in Neighborhood Perception. *Political Analysis*, *34*(3), 479–487. https://doi.org/10.1017/pan.2025.10022

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, *21*(1), C1–C68. https://doi.org/10.1111/ectj.12097

7.1 参考文献 II

Chew, R., Bollenbacher, J., Wenger, M., Speer, J., & Kim, A. (2023, June 23). *LLM-Assisted Content Analysis: Using Large Language Models to Support Deductive Coding*. arXiv: 2306.14924 [cs.CL]. https://doi.org/10.48550/arXiv.2306.14924

Dennstädt, F., Zink, J., Putora, P. M., Hastings, J., & Cihoric, N. (2024). Title and abstract screening for literature reviews using large language models: An exploratory study in the biomedical domain. *Systematic Reviews*, *13*(1), 158. https://doi.org/10.1186/s13643-024-02575-4

Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). Can AI language models replace human participants? *Trends in Cognitive Sciences*, *27*(7), 597–600. https://doi.org/10.1016/j.tics.2023.04.008

Dominguez-Olmedo, R., Hardt, M., & Mendler-Dünner, C. (2024, December 9). *Questioning the Survey Responses of Large Language Models*. arXiv: 2306.07951 [cs.CL]. https://doi.org/10.48550/arXiv.2306.07951

Egami, N., Hinck, M., Stewart, B. M., & Wei, H. (2024). Using large language model annotations for the social sciences: A general framework of using predicted variables in downstream analyses. *Preprint from November*, *17*, 2024. Retrieved May 12, 2026, from https://naokiegami.com/paper/dsl_ss.pdf

Feng, S., Wan, H., Wang, N., Tan, Z., Luo, M., & Tsvetkov, Y. (2024). What Does the Bot Say? Opportunities and Risks of Large Language Models in Social Media Bot Detection. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 3580–3601). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.196

Gao, C., Lan, X., Lu, Z., Mao, J., Piao, J., Wang, H., Jin, D., & Li, Y. (2025, June 3). *S^3: Social-network Simulation System with Large Language Model-Empowered Agents*. arXiv: 2307.14984 [cs.SI]. https://doi.org/10.48550/arXiv.2307.14984

Gehrmann, J., Quakulinski, L., & Beyan, O. (2024). Large Language Models for Literature Reviews - An Exemplary Comparison of LLM-based Approaches with Manual Methods. *2024 2nd International Conference on Foundation and Large Language Models (FLLM)*, 385–391. https://doi.org/10.1109/FLLM63129.2024.10852447

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120

Goyal, T., Li, J. J., & Durrett, G. (2023, May 23). *News Summarization and Evaluation in the Era of GPT-3*. arXiv: 2209.12356 [cs.CL]. https://doi.org/10.48550/arXiv.2209.12356

Grimmer, J., Roberts, M. E., & Stewart, B. M. (2021). Machine Learning for Social Science: An Agnostic Approach. *Annual Review of Political Science*, *24*, 395–419. https://doi.org/10.1146/annurev-polisci-053119-015921

7.1 参考文献 III

Hackenburg, K., & Margetts, H. (2024). Evaluating the persuasive influence of political microtargeting with large language models. *Proceedings of the National Academy of Sciences of the United States of America*, *121*(24), e2403116121. https://doi.org/10.1073/pnas.2403116121

Hai Liang. (2026). *A Computational Approach to Human - AI Communication*. Singapore. www.dropbox.com/scl/fi/nlq3k73akoln87ebkkxsr/NUS.pdf?rlkey=0kzml5htrvthlhvqa6m6w9ee2&e=1&dl=0

Hämäläinen, P., Tavast, M., & Kunnari, A. (2023). Evaluating Large Language Models in Generating Synthetic HCI Research Data: A Case Study. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 1–19. https://doi.org/10.1145/3544548.3580688

Hanschmann, L., Mokelke, M., & Maedche, A. (2025). LadderChat An LLM-Based Conversational Agent for Laddering Interviews. In A. Følstad, S. Papadopoulos, T. Araujo, E. L.-C. Law, E. Luger, S. Hobert, & P. B. Brandtzaeg (Eds.), *Chatbots and Human-Centered AI* (pp. 48–65). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-88045-2_4

Jiang, Y., Qiu, R., Zhang, Y., & Zhang, P.-F. (2024). Balanced and Explainable Social Media Analysis for Public Health with Large Language Models. In Z. Bao, R. Borovica-Gajic, R. Qiu, F. Choudhury, & Z. Yang (Eds.), *Databases Theory and Applications* (pp. 73–86). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-47843-7_6

Ke, P., Wen, B., Feng, A., Liu, X., Lei, X., Cheng, J., Wang, S., Zeng, A., Dong, Y., Wang, H., Tang, J., & Huang, M. (2024). CritiqueLLM: Towards an Informative Critique Generation Model for Evaluation of Large Language Model Generation. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 13034–13054). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.704

King, G., & Worthington, S. (2026). AI Article Reviews Dominate Human Peer Reviews at Their Best: Evidence and Policy Choices. Retrieved November 4, 2026, from https://gking.harvard.edu/peerless/

Kolluri, A., Wu, S., Park, J. S., & Bernstein, M. S. (2025). Finetuning LLMs for Human Behavior Prediction in Social Science Experiments. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (pp. 30096–30111). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-main.1530

Li, H., Yu, J., Li, R., Hao, Z., Xuan, Y., Yuan, J., Xu, B., Li, J., & Liu, Z. (2024). LM-Interview: An Easy-to-use Smart Interviewer System via Knowledge-guided Language Model Exploitation. In D. I. Hernandez Farias, T. Hope, & M. Li (Eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 520–528). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-demo.52

7.1 参考文献 IV

Li, M., Lau, J. H., & Hovy, E. (2024). A Sentiment Consolidation Framework for Meta-Review Generation. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 10158–10177). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.547

Liang, W., Zhang, Y., Cao, H., Wang, B., Ding, D. Y., Yang, X., Vodrahalli, K., He, S., Smith, D. S., Yin, Y., McFarland, D. A., & Zou, J. (2024). Can Large Language Models Provide Useful Feedback on Research Papers? A Large-Scale Empirical Analysis. *NEJM AI*, *1*(8), Aloa2400196. https://doi.org/10.1056/Aloa2400196

Lin, Y.-T., & Chen, Y.-N. (2023). LLM-Eval: Unified Multi-Dimensional Automatic Evaluation for Open-Domain Conversations with Large Language Models. In Y.-N. Chen & A. Rastogi (Eds.), *Proceedings of the 5th Workshop on NLP for Conversational AI (NLP4ConvAI 2023)* (pp. 47–58). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.nlp4convai-1.5

Liu, A. Y., & Zhang, M. M. (2026). Positivity Bias in AI-Generated Summaries of User-Generated Content: Exploring Its Sources and Impact on Public Sentiment. *Computational Communication Research*, *8*(2), 1. https://doi.org/10.5117/CCR2026.2.4.LIU

Luceri, L., Boniardi, E., & Ferrara, E. (2024). Leveraging Large Language Models to Detect Influence Campaigns on Social Media. *Companion Proceedings of the ACM Web Conference 2024*, 1459–1467. https://doi.org/10.1145/3589335.3651912

Manning, B. S., Zhu, K., & Horton, J. J. (2024, April). *Automated Social Science: Language Models as Scientist and Subjects*. 32381. https://doi.org/10.3386/w32381

Ouyang, R., Chu, C., Xin, Z., & Ma, X. (2025). PDFMathTranslate: Scientific document translation preserving layouts. In I. Habernal, P. Schulam, & J. Tiedemann (Eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 918–924). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-demos.71

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative Agents: Interactive Simulacra of Human Behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 1–22. https://doi.org/10.1145/3586183.3606763

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023, August 6). *Generative Agents: Interactive Simulacra of Human Behavior*. arXiv: 2304.03442 [cs.HC]. https://doi.org/10.48550/arXiv.2304.03442

Park, J. S., Zou, C. Q., Kamphorst, J., Egan, N., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Liang, P., Willer, R., & Bernstein, M. S. (2026, June 28). *LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals*. arXiv: 2411.10109 [cs.AI]. https://doi.org/10.48550/arXiv.2411.10109

7.1 参考文献 V

Pataranutaporn, P., Powdthavee, N., Archiwaranguprok, C., & Maes, P. (2025). Simulating human well-being with large language models: Systematic validation and misestimation across 64,000 individuals from 64 countries. *Proceedings of the National Academy of Sciences*, *122*(48), e2519394122. https://doi.org/10.1073/pnas.2519394122

Poole, D. L., & Mackworth, A. K. (2023, July 13). *Artificial Intelligence: Foundations of Computational Agents*. Cambridge University Press. https://doi.org/10.1017/9781009258227

Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose Opinions Do Language Models Reflect? *Proceedings of the 40th International Conference on Machine Learning*, 29971–30004. Retrieved November 3, 2026, from https://proceedings.mlr.press/v202/santurkar23a.html

Shankar, S., Zamfirescu-Pereira, J., Hartmann, B., Parameswaran, A., & Arawjo, I. (2024). Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences. *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 1–14. https://doi.org/10.1145/3654777.3676450

Si, C., Yang, D., & Hashimoto, T. (2024, September 6). *Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers*. arXiv: 2409.04109 [cs.CL]. https://doi.org/10.48550/arXiv.2409.04109

Tao, Y., Viberg, O., Baker, R. S., & Kizilcec, R. F. (2024). Cultural bias and cultural alignment of large language models (M. Muthukrishna, Ed.). *PNAS Nexus*, *3*(9), pgae346. https://doi.org/10.1093/pnasnexus/pgae346

Tuggener, D., Schneider, T., Huwiler, A., Kreienbühl, T., Hischier, S., von Däniken, P., & Niehaus, S. (2024). Role-Playing LLMs in Professional Communication Training: The Case of Investigative Interviews with Children. In P. H. Luz de Araujo, A. Baumann, D. Gromann, B. Krenn, B. Roth, & M. Wiegand (Eds.), *Proceedings of the 20th Conference on Natural Language Processing (KONVENS 2024)* (pp. 249–263). Association for Computational Linguistics. Retrieved November 4, 2026, from https://aclanthology.org/2024.konvens-main.26/

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*. https://papers.nips.cc/paper/7181-attention-is-all-you-need

Velez, Y. R., & Liu, P. (2025). Confronting Core Issues: A Critical Assessment of Attitude Polarization Using Tailored Experiments. *American Political Science Review*, *119*(2), 1036–1053. https://doi.org/10.1017/S0003055424000819

Wang, A., Morgenstern, J., & Dickerson, J. P. (2025). Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence*, *7*(3), 400–411. https://doi.org/10.1038/s42256-025-00986-z

7.1 参考文献 VI

Wang, L., Zhang, J., Yang, H., Chen, Z.-Y., Tang, J., Zhang, Z., Chen, X., Lin, Y., Sun, H., Song, R., Zhao, X., Xu, J., Dou, Z., Wang, J., & Wen, J.-R. (2025). User Behavior Simulation with Large Language Model-based Agents. *ACM Transactions on Information Systems*, *43*(2), 55:1–55:37. https://doi.org/10.1145/3708985

Wang, Q., Downey, D., Ji, H., & Hope, T. (2024). SciMON: Scientific Inspiration Machines Optimized for Novelty. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 279–299). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.18

Wang, X., Kim, H., Rahman, S., Mitra, K., & Miao, Z. (2024). Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 1–21. https://doi.org/10.1145/3613904.3641960

Xie, C., Chen, C., Jia, F., Ye, Z., Lai, S., Shu, K., Gu, J., Bibi, A., Hu, Z., Jurgens, D., Evans, J., Torr, P., Ghanem, B., & Li, G. (2024, November 1). *Can Large Language Model Agents Simulate Human Trust Behavior?* arXiv: 2402.04559 [cs.AI]. https://doi.org/10.48550/arXiv.2402.04559

Xie, Y., & Xie, Y. (2025, March 2). *Variance reduction in output from generative AI*. arXiv: 2503.01033 [cs.CY]. https://doi.org/10.48550/arXiv.2503.01033

Xu, F., Shao, C., Chen, L., Zeng, Q., Chen, Z., Sukiennik, N., Wang, J., Gao, C., Wang, H., Lian, J., Meng, Q., Xie, X., Li, Y., & Evans, J. (2026). Large language models for computational social science. *Nature Human Behaviour*. https://doi.org/10.1038/s41562-026-02539-9

Yan, Y., Zeng, Q., Zheng, Z., Yuan, J., Feng, J., Zhang, J., Xu, F., & Li, Y. (2024, October 11). *OpenCity: A Scalable Platform to Simulate Urban Activities with Massive LLM Agents*. arXiv: 2410.21286 [cs.MA]. https://doi.org/10.48550/arXiv.2410.21286

Yang, Z., Du, X., Li, J., Zheng, J., Poria, S., & Cambria, E. (2024). Large Language Models for Automated Open-domain Scientific Hypotheses Discovery. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the Association for Computational Linguistics: ACL 2024* (pp. 13545–13565). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-acl.804

Yu, J., Ren, Y., Gong, C., Tan, J., Li, X., & Zhang, X. (2025). Leveraging Large Language Models for Node Generation in Few-Shot Learning on Text-Attributed Graphs. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(12), 13087–13095. https://doi.org/10.1609/aaai.v39i12.33428

Zhang, J., Xu, X., Zhang, N., Liu, R., Hooi, B., & Deng, S. (2024). Exploring Collaboration Mechanisms for LLM Agents: A Social Psychology View. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 14544–14607). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.782

7.1 参考文献 VII

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023, December 24). *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. arXiv: 2306.05685 [cs.CL]. https://doi.org/10.48550/arXiv.2306.05685

Zhou, A., Mukerjee, S., Maslowska, E., Duan, Z., Jaidka, K., Lukito, J., Shugars, S., Smith, M. A., Trilling, D., & Yang, T. (2026). Be Careful What You Prompt For: Generative AI in Computational Communication Research. *Computational Communication Research*, *8*(2), 1–34. https://doi.org/10.5117/CCR2026.2.1.ZHOU